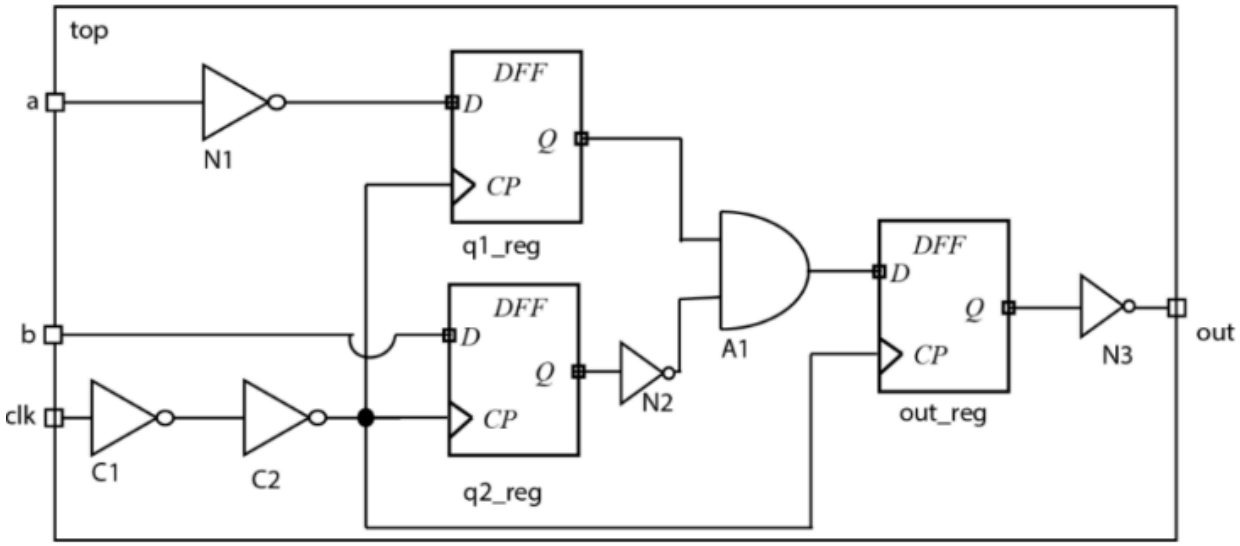
VDF A2 REPORT

Nishchal

2022330

Top Module



Q1 cadence/tempus :

```
[nishchal22330@edatools-server2 nish_vdf]$ csh
[nishchal22330@edatools-server2 nish_vdf]$ source /cadence/cshrc


Welcome to Cadence Tools Suite

[nishchal22330@edatools-server2 nish_vdf]$ tempus -file STA.tcl

Cadence Tempus(TM) Timing Signoff Solution.
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Version:        v20.10-p003_1, built Wed Apr 29 15:56:37 PDT 2020
Options:        -file STA.tcl
Date:           Tue Oct 28 15:30:29 2025
Host:           edatools-server2.iiitd.edu.in (x86_64 w/Linux 2.6.32-754.35.1.el
6.x86_64) (18cores*72cpus*Intel(R) Xeon(R) Gold 6354 CPU @ 3.00GHz 39936KB)
OS:             Red Hat Enterprise Linux Server release 6.10 (Santiago)

License:
                tpsxl   Tempus Timing Signoff Solution XL       20.1    checkout
 succeeded
                16 CPU jobs allowed with the current license(s). Use set_multi_c
pu_usage to set your required CPU count.
Change the soft stacksize limit to 0.2%RAM (1033 mbytes). Set global soft_stack_
size_limit to change the value.
environment variable TMPDIR is '/tmp/ssv_tmpdir_71305_kslPIt'.
#@ Processing -files option
Sourcing tcl/tk file 'STA.tcl' ...
Scheduling  timing library file(s) 'slow.lib' to be loaded when the set_top_modu
le command is issued
Scheduling netlist file(s) '/home/harshit22210/Desktop/nish/syn_net.v' to be loa
ded when the set_top_module command is issued
#% Begin Load MMMC data ... (date=10/28 15:30:38, mem=493.3M)
Number of views requested 1 are allowed with the current licenses... continuing
with set_analysis_view.
#% End Load MMMC data ... (date=10/28 15:30:38, total cpu=0:00:00.0, real=0:00:0
0.0, peak res=493.4M, current mem=493.4M)
Loading view definition file from .ssv_emulate_view_definition_71305.tcl
Reading default_emulate_libset_max timing library '/home/nishchal22330/Desktop/c
```

Q2 top.v (netlist)

```
`timescale 1ns / 1ps

module syn_netlist_top (
    input  wire a,
    input  wire b,
    input  wire clk,
    output wire out
);

    wire a_n1;         // output of N1
    wire clk_c1;       // after C1
    wire clk_del;      // after C2 (delayed clock)
    wire q1_q;         // q1_reg Q
    wire q2_q;         // q2_reg Q
    wire q2_q_n;       // inverted q2_q (N2)
    wire a1_out;       // output of OR gate A1 (D for out_reg)
    wire out_q;        // out_reg Q
    // ----- Clock inverters (C1, C2) -----
    // clk -> C1 -> clk_c1 -> C2 -> clk_del
    INVX1 u_C1 ( .A(clk),     .Y(clk_c1) );
    INVX1 u_C2 ( .A(clk_c1),  .Y(clk_del) );

    // ----- Input inverter N1 for 'a' -----
    INVX1 u_N1 ( .A(a), .Y(a_n1) );

    // ----- D flip-flops (positive-edge) -----
    // q1_reg captures a_n1 on rising edge of clk_del
    DFFX1 u_q1_reg ( .D(a_n1), .CK(clk_del), .Q(q1_q) );

    // q2_reg captures b on rising edge of clk_del
    DFFX1 u_q2_reg ( .D(b), .CK(clk_del), .Q(q2_q) );

    // out_reg captures combinational A1 result on rising edge of clk (main clock)
    DFFX1 u_out_reg ( .D(a1_out), .CK(clk), .Q(out_q) );

    // ----- Inverter N2 (inverts q2_q before OR) -----
    INVX1 u_N2 ( .A(q2_q), .Y(q2_q_n) );

    // ----- OR gate A1 (q1_q OR (not q2_q)) -----
    OR2X1 u_A1 ( .A(q1_q), .B(q2_q_n), .Y(a1_out) );

    // ----- Output inverter N3 -----
    INVX1 u_N3 ( .A(out_q), .Y(out) );

endmodule
```

Q3 top.sdc + Timing Report

sdc:

```
create_clock -name clk -period 10 [get_ports clk]

set_input_delay 2 -clock clk [get_ports {a b}]

set_output_delay 2 -clock clk [get_ports out]

set_drive 0 [get_ports {a b clk}]

set_load 0.1 [get_ports out]
```

Worst Setup :

```
#====================================================================
Path 1: MET Late External Delay Assertion
Endpoint:   out         (^) checked with  leading edge of 'clk'
Beginpoint: u_out_reg/Q (v) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- External Delay                2.000
+ Phase Shift                  10.000
= Required Time                 8.000
- Arrival Time                  0.924
= Slack Time                    7.076
     Clock Rise Edge                 0.000
     + Clock Network Latency (Ideal) 0.000
     = Beginpoint Arrival Time       0.000
      ------------------------------------------------------------
      Instance   Arc          Cell   Delay  Arrival  Required
                                            Time     Time
      ------------------------------------------------------------
      u_out_reg  CK ^         -      -      0.000    7.076
      u_out_reg  CK ^ -> Q v  DFFX1  0.340  0.340    7.416
      u_N3       A v -> Y ^   INVX1  0.584  0.924    8.000
      -          out ^        -      0.000  0.924    8.000
      ------------------------------------------------------------
```

Worst hold:

```
Path 1: MET Hold Check with Pin u_out_reg/CK
Endpoint:   u_out_reg/D (^) checked with  leading edge of 'clk'
Beginpoint: u_q1_reg/Q  (^) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
+ Hold                          0.011
+ Phase Shift                   0.000
= Required Time                 0.011
  Arrival Time                  0.363
  Slack Time                    0.352
     Clock Rise Edge                     0.000
     + Clock Network Latency (Ideal)     0.000
     = Beginpoint Arrival Time           0.000
      ------------------------------------------------------------
      Instance   Arc          Cell   Delay  Arrival  Required
                                            Time     Time
      ------------------------------------------------------------
      u_q1_reg   CK ^         -      -      0.000    -0.352
      u_q1_reg   CK ^ -> Q ^  DFFX1  0.275  0.275    -0.077
      u_A1       A ^ -> Y ^   OR2X1  0.088  0.363    0.011
      u_out_reg  D ^          DFFX1  0.000  0.363    0.011
      ------------------------------------------------------------
```

Q4 clock uncertainty + latency Timing Report

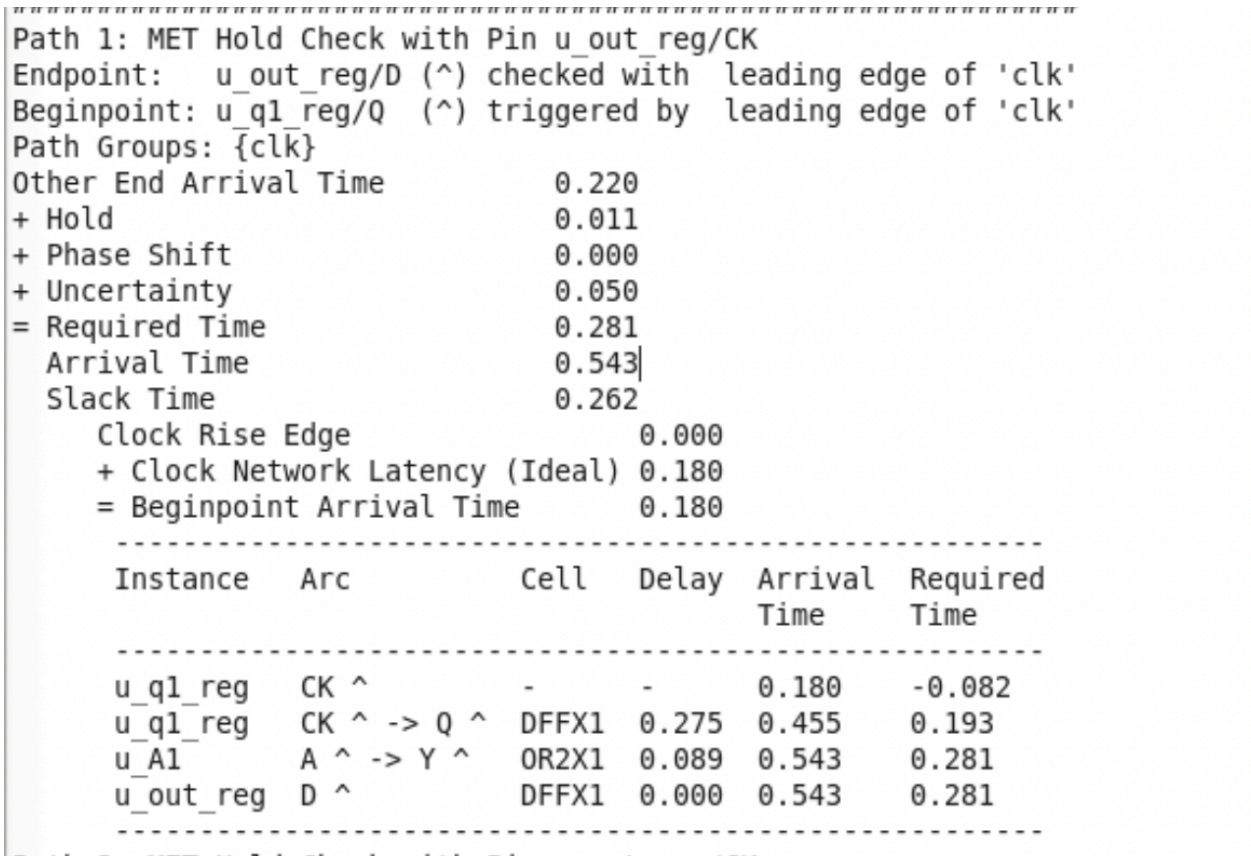
sdc:

```
create_clock -name clk -period 10 [get_ports clk]
set_clock_latency -source -early 0.18  [get_clocks clk]
set_clock_latency -source -late  0.22  [get_clocks clk]

set_clock_latency -early 0.75  [get_clocks clk]
set_clock_latency -late  0.85  [get_clocks clk]


set_clock_uncertainty -setup 0.10 [get_clocks clk]
set_clock_uncertainty -hold  0.05 [get_clocks clk]

set_input_delay 2 -clock clk [get_ports {a b}]

set_output_delay 2 -clock clk [get_ports out]

set_drive 0 [get_ports {a b clk}]

set_load 0.1 [get_ports out]
```

Worst Setup :

```
Path 1: MET Late External Delay Assertion
Endpoint:   out         (^) checked with  leading edge of 'clk'
Beginpoint: u_out_reg/Q (v) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
+ Source Insertion Delay        0.180
- External Delay                2.000
+ Phase Shift                  10.000
- Uncertainty                   0.100
= Required Time                 8.080
- Arrival Time                  1.144
= Slack Time                    6.936
     Clock Rise Edge                 0.000
     + Clock Network Latency (Ideal) 0.220
     = Beginpoint Arrival Time       0.220
      ------------------------------------------------------------
      Instance   Arc          Cell   Delay  Arrival  Required
                                            Time     Time
      ------------------------------------------------------------
      u_out_reg  CK ^         -      -      0.220    7.156
      u_out_reg  CK ^ -> Q v  DFFX1  0.340  0.560    7.496
      u_N3       A v -> Y ^   INVX1  0.584  1.144    8.080
      -          out ^        -      0.000  1.144    8.080
      ------------------------------------------------------------
```

Worst hold:



```
****************************************************************
Path 1: MET Hold Check with Pin u_out_reg/CK
Endpoint:   u_out_reg/D (^) checked with  leading edge of 'clk'
Beginpoint: u_q1_reg/Q  (^) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.220
+ Hold                          0.011
+ Phase Shift                   0.000
+ Uncertainty                   0.050
= Required Time                 0.281
  Arrival Time                  0.543
  Slack Time                    0.262
     Clock Rise Edge                     0.000
     + Clock Network Latency (Ideal)     0.180
     = Beginpoint Arrival Time           0.180
      -------------------------------------------------------
      Instance   Arc          Cell   Delay  Arrival  Required
                                            Time     Time
      -------------------------------------------------------
      u_q1_reg   CK ^         -      -      0.180    -0.082
      u_q1_reg   CK ^ -> Q ^  DFFX1  0.275  0.455    0.193
      u_A1       A ^ -> Y ^   OR2X1  0.089  0.543    0.281
      u_out_reg  D ^          DFFX1  0.000  0.543    0.281
      -------------------------------------------------------
```

Adding realistic clock latency and uncertainty tightened the timing on the same data path. The data delay is unchanged (~0.924 ns), but introducing 0.22 ns launch clock latency pushes the data arrival later (0.924 → 1.144 ns). The setup requirement also shrinks slightly due to 0.10 ns setup uncertainty (Required 8.000 → 8.080 ns). Net effect: worst-path setup slack reduces from **7.076 ns** (original) to **6.936 ns** (new), i.e., **–0.140 ns**. For hold, adding early clock latency and 0.05 ns hold uncertainty tightens the check window, dropping worst-path hold slack from **0.352 ns** to **0.262 ns** (**–0.090 ns**). The path still meets timing, but with less margin after modeling clock effects.

Q5 Double clock Frequency

sdc:

```
create_clock -name clk -period 5 [get_ports clk]

set_input_delay 2 -clock clk [get_ports {a b}]

set_output_delay 2 -clock clk [get_ports out]

set_drive 0 [get_ports {a b clk}]

set_load 0.1 [get_ports out]
```

Worst Setup :

```
Path 1: MET Late External Delay Assertion
Endpoint:   out         (^) checked with  leading edge of 'clk'
Beginpoint: u_out_reg/Q (v) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- External Delay                2.000
+ Phase Shift                   5.000
= Required Time                 3.000
- Arrival Time                  0.924
= Slack Time                    2.076
     Clock Rise Edge                 0.000
     + Clock Network Latency (Ideal) 0.000
     = Beginpoint Arrival Time       0.000
      ------------------------------------------------------------
      Instance   Arc          Cell   Delay  Arrival  Required
                                            Time     Time
      ------------------------------------------------------------
      u_out_reg  CK ^         -      -      0.000    2.076
      u_out_reg  CK ^ -> Q v  DFFX1  0.340  0.340    2.416
      u_N3       A v -> Y ^   INVX1  0.584  0.924    3.000
      -          out ^        -      0.000  0.924    3.000
      ------------------------------------------------------------
```

Worst hold:

```
Path 1: MET Hold Check with Pin u_out_reg/CK
Endpoint:   u_out_reg/D (^) checked with  leading edge of 'clk'
Beginpoint: u_q1_reg/Q  (^) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
+ Hold                          0.011
+ Phase Shift                   0.000
= Required Time                 0.011
  Arrival Time                  0.363
  Slack Time                    0.352
     Clock Rise Edge                     0.000
     + Clock Network Latency (Ideal)     0.000
     = Beginpoint Arrival Time           0.000
      -------------------------------------------------------
      Instance   Arc          Cell   Delay  Arrival  Required
                                            Time     Time
      -------------------------------------------------------
      u_q1_reg   CK ^         -      -      0.000    -0.352
      u_q1_reg   CK ^ -> Q ^  DFFX1  0.275  0.275    -0.077
      u_A1       A ^ -> Y ^   OR2X1  0.088  0.363    0.011
      u_out_reg  D ^          DFFX1  0.000  0.363    0.011
      -------------------------------------------------------
```

Cutting the clock period from 10 ns to 5 ns squeezed only the setup window. The data path delay stayed ~0.924 ns, but the required time shrank from 8.000 ns to 3.000 ns (phase-shift equals period, minus the same 2 ns external delay). So setup slack dropped from **7.076 ns** → **2.076 ns** (a **–5.000 ns** change), making the same path much closer to critical but still meeting timing. The **hold slack is unchanged at ~0.352 ns** because hold is a same-edge check; with the same early clock, uncertainty, and data path, tightening the period doesn't affect hold.

Q6 Tight Input Delay

sdc:

```
create_clock -name clk -period 10 [get_ports clk]

set_input_delay -max 4 -clock clk [get_ports {a b}]
set_input_delay -min 0.2 -clock clk [get_ports {a b}]

set_output_delay 2 -clock clk [get_ports out]

set_drive 0 [get_ports {a b clk}]

set_load 0.1 [get_ports out]
```

Setup of all paths :

```
Path 1: MET Setup Check with Pin u_q1_reg/CK
Endpoint:   u_q1_reg/D (^) checked with  leading edge of 'clk'
Beginpoint: a          (v) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- Setup                         0.186
+ Phase Shift                  10.000
= Required Time                 9.814
- Arrival Time                  4.025
= Slack Time                    5.789
     Clock Rise Edge                         0.000
     + Input Delay                           4.000
     = Beginpoint Arrival Time               4.000
      ------------------------------------------------------
      Instance  Arc         Cell   Delay  Arrival  Required
                                          Time     Time
      ------------------------------------------------------
      -         a v         -      -      4.000    9.789
      u_N1      A v -> Y ^  INVX1  0.025  4.025    9.814
      u_q1_reg  D ^         DFFX1  0.000  4.025    9.814
      ------------------------------------------------------
```

```
Path 2: MET Setup Check with Pin u_q2_reg/CK
Endpoint:   u_q2_reg/D (^) checked with  leading edge of 'clk'
Beginpoint: b          (^) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- Setup                         0.202
+ Phase Shift                  10.000
= Required Time                 9.798
- Arrival Time                  4.000
= Slack Time                    5.798
     Clock Rise Edge                        0.000
     + Input Delay                          4.000
     = Beginpoint Arrival Time              4.000
      ---------------------------------------------
      Instance  Arc  Cell   Delay  Arrival  Required
                                    Time     Time
      ---------------------------------------------
      -         b ^  -      -      4.000    9.798
      u_q2_reg  D ^  DFFX1  0.000  4.000    9.798
      ---------------------------------------------
```

```
Path 3: MET Setup Check with Pin u_q1_reg/CK
Endpoint:   u_q1_reg/D (v) checked with  leading edge of 'clk'
Beginpoint: a          (^) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- Setup                         0.096
+ Phase Shift                  10.000
= Required Time                 9.904
- Arrival Time                  4.021
= Slack Time                    5.883
     Clock Rise Edge                        0.000
     + Input Delay                          4.000
     = Beginpoint Arrival Time              4.000
      -----------------------------------------------------
      Instance  Arc         Cell   Delay  Arrival  Required
                                          Time     Time
      -----------------------------------------------------
      -         a ^         -      -      4.000    9.883
      u_N1      A ^ -> Y v  INVX1  0.021  4.021    9.904
      u_q1_reg  D v         DFFX1  0.000  4.021    9.904
      -----------------------------------------------------
```

```
Path 4: MET Setup Check with Pin u_q2_reg/CK
Endpoint:   u_q2_reg/D (v) checked with  leading edge of 'clk'
Beginpoint: b          (v) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- Setup                         0.109
+ Phase Shift                  10.000
= Required Time                 9.891
- Arrival Time                  4.000
= Slack Time                    5.891
     Clock Rise Edge                      0.000
     + Input Delay                        4.000
     = Beginpoint Arrival Time            4.000
      ------------------------------------------------
      Instance  Arc  Cell   Delay  Arrival  Required
                                    Time     Time
      ------------------------------------------------
      -         b v  -      -      4.000    9.891
      u_q2_reg  D v  DFFX1  0.000  4.000    9.891
      ------------------------------------------------
```

```
Path 5: MET Late External Delay Assertion
Endpoint:   out         (^) checked with  leading edge of 'clk'
Beginpoint: u_out_reg/Q (v) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- External Delay                2.000
+ Phase Shift                  10.000
= Required Time                 8.000
- Arrival Time                  0.924
= Slack Time                    7.076
     Clock Rise Edge                     0.000
     + Clock Network Latency (Ideal)     0.000
     = Beginpoint Arrival Time           0.000
      ------------------------------------------------------
      Instance   Arc          Cell   Delay  Arrival  Required
                                            Time     Time
      ------------------------------------------------------
      u_out_reg  CK ^         -      -      0.000    7.076
      u_out_reg  CK ^ -> Q v  DFFX1  0.340  0.340    7.416
      u_N3       A v -> Y ^   INVX1  0.584  0.924    8.000
      -          out ^        -      0.000  0.924    8.000
      ------------------------------------------------------
```

Effect of Tighter Input Delay on Setup Slack:

Tightening the **input-delay (setup)** on ports `a,b` from **2 ns → 4 ns** eats directly into the launch time at the input flops, so every setup path that starts at an input now arrives **2 ns later** and loses exactly **2 ns** of margin. In your reports the input paths show:

- via `a → u_q1_reg/D`: slack **5.789 ns** (Required 9.814, Arrival 4.025). With the old 2 ns delay this would have been ≈ **7.789 ns**.
- via `b → u_q2_reg/D`: slack **5.798 ns** (Req 9.798, Arr 4.000) vs ≈ **7.798 ns** before.
- via `a → u_q1_reg/D` (alt path): slack **5.883 ns** (Req 9.904, Arr 4.021) vs ≈ **7.883 ns** before.

So, making the setup constraint tighter on the inputs reduces setup slack on all input-originating paths by ~**2.0 ns**, while non-input paths (e.g., the output path with 7.076 ns slack) are unaffected.

Q7 Tight output Delay

sdc:

```
create_clock -name clk -period 10 [get_ports clk]

set_input_delay  2 -clock clk [get_ports {a b}]

set_output_delay -max 1.0 -clock clk [get_ports out]

set_drive 0 [get_ports {a b clk}]

set_load 0.1 [get_ports out]
```

Setup for all paths :

```
Path 1: MET Setup Check with Pin u_q1_reg/CK
Endpoint:   u_q1_reg/D (^) checked with  leading edge of 'clk'
Beginpoint: a          (v) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- Setup                         0.186
+ Phase Shift                  10.000
= Required Time                 9.814
- Arrival Time                  2.025
= Slack Time                    7.789
     Clock Rise Edge                      0.000
     + Input Delay                        2.000
     = Beginpoint Arrival Time            2.000
      ------------------------------------------------------
      Instance  Arc         Cell   Delay  Arrival  Required
                                          Time     Time
      ------------------------------------------------------
      -         a v         -      -      2.000    9.789
      u_N1      A v -> Y ^  INVX1  0.025  2.025    9.814
      u_q1_reg  D ^         DFFX1  0.000  2.025    9.814
      ------------------------------------------------------
```

```
      -------------------------------------------------------
Path 2: MET Setup Check with Pin u_q2_reg/CK
Endpoint:   u_q2_reg/D (^) checked with  leading edge of 'clk'
Beginpoint: b          (^) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- Setup                         0.202
+ Phase Shift                  10.000
= Required Time                 9.798
- Arrival Time                  2.000
= Slack Time                    7.798
     Clock Rise Edge                      0.000
     + Input Delay                        2.000
     = Beginpoint Arrival Time            2.000
      ---------------------------------------------
      Instance  Arc  Cell   Delay  Arrival  Required
                                   Time     Time
      ---------------------------------------------
      -         b ^  -      -      2.000    9.798
      u_q2_reg  D ^  DFFX1  0.000  2.000    9.798
      ---------------------------------------------
```

```
Path 3: MET Setup Check with Pin u_q1_reg/CK
Endpoint:   u_q1_reg/D (v) checked with  leading edge of 'clk'
Beginpoint: a          (^) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- Setup                         0.096
+ Phase Shift                  10.000
= Required Time                 9.904
- Arrival Time                  2.021
= Slack Time                    7.883
     Clock Rise Edge                      0.000
     + Input Delay                        2.000
     = Beginpoint Arrival Time            2.000
      -------------------------------------------------------
      Instance  Arc         Cell   Delay  Arrival  Required
                                          Time     Time
      -------------------------------------------------------
      -         a ^         -      -      2.000    9.883
      u_N1      A ^ -> Y v  INVX1  0.021  2.021    9.904
      u_q1_reg  D v         DFFX1  0.000  2.021    9.904
      -------------------------------------------------------
```

```
Path 4: MET Setup Check with Pin u_q2_reg/CK
Endpoint:   u_q2_reg/D (v) checked with  leading edge of 'clk'
Beginpoint: b          (v) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- Setup                         0.109
+ Phase Shift                  10.000
= Required Time                 9.891
- Arrival Time                  2.000
= Slack Time                    7.891
     Clock Rise Edge                      0.000
     + Input Delay                        2.000
     = Beginpoint Arrival Time            2.000
      ---------------------------------------------
      Instance  Arc  Cell   Delay  Arrival  Required
                                    Time     Time
      ---------------------------------------------
      -         b v  -      -      2.000    9.891
      u_q2_reg  D v  DFFX1  0.000  2.000    9.891
      ---------------------------------------------
```

```
Path 5: MET Late External Delay Assertion
Endpoint:   out         (^) checked with  leading edge of 'clk'
Beginpoint: u_out_reg/Q (v) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- External Delay                1.000
+ Phase Shift                  10.000
= Required Time                 9.000
- Arrival Time                  0.924
= Slack Time                    8.076
     Clock Rise Edge                 0.000
     + Clock Network Latency (Ideal) 0.000
     = Beginpoint Arrival Time       0.000
      -----------------------------------------------------
      Instance   Arc          Cell   Delay  Arrival  Required
                                            Time     Time
      -----------------------------------------------------
      u_out_reg  CK ^         -      -      0.000    8.076
      u_out_reg  CK ^ -> Q v  DFFX1  0.340  0.340    8.416
      u_N3       A v -> Y ^   INVX1  0.584  0.924    9.000
      -          out ^        -      0.000  0.924    9.000
      -----------------------------------------------------
```

Tightening the **output-delay (setup)** on `out` from **2 ns → 1 ns** gives your block **more of the clock period** to produce the signal before it leaves the chip. In the timing report, the required time at `out` moves from **8.000 ns** (with 2 ns external delay) to **9.000 ns** (with 1 ns), while the path's data arrival stays the same at **0.924 ns**. Therefore the worst output path's setup slack improves from **7.076 ns → 8.076 ns** (**+1.000 ns**). Only output-terminating paths change; input and internal reg-to-reg paths are unaffected.

Q8

Add multi-cycle path exception for ONLY the setup analysis with a path multiplier of 4

on the path showing the worst setup slack.

sdc:

```
create_clock -name clk -period 10 [get_ports clk]

set_input_delay  2 -clock clk [get_ports {a b}]

set_output_delay 2 -clock clk [get_ports out]

set_multicycle_path 4 -setup -from [get_pins -hierarchical {u_q1_reg/Q} ] -to [get_ports {out}]
set_drive 0 [get_ports {a b clk}]

set_load 0.1 [get_ports out]
```

Worst Setup :

```
Path 1: MET Late External Delay Assertion
Endpoint:   out         (^) checked with  leading edge of 'clk'
Beginpoint: u_out_reg/Q (v) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- External Delay                2.000
+ Phase Shift                  10.000
+ Cycle Adjustment             30.000
= Required Time                38.000
- Arrival Time                  0.924
= Slack Time                   37.076
     Clock Rise Edge                     0.000
     + Clock Network Latency (Ideal)     0.000
     = Beginpoint Arrival Time           0.000
      ------------------------------------------------------------
      Instance   Arc          Cell   Delay  Arrival  Required
                                            Time     Time
      ------------------------------------------------------------
      u_out_reg  CK ^         -      -      0.000    37.076
      u_out_reg  CK ^ -> Q v  DFFX1  0.340  0.340    37.416
      u_N3       A v -> Y ^   INVX1  0.584  0.924    38.000
      -          out ^        -      0.000  0.924    38.000
      ------------------------------------------------------------
```

Worst hold:

```
Path 1: MET Hold Check with Pin u_out_reg/CK
Endpoint:   u_out_reg/D (^) checked with  leading edge of 'clk'
Beginpoint: u_q1_reg/Q  (^) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
+ Hold                          0.011
+ Phase Shift                   0.000
= Required Time                 0.011
  Arrival Time                  0.363
  Slack Time                    0.352
     Clock Rise Edge                     0.000
     + Clock Network Latency (Ideal)     0.000
     = Beginpoint Arrival Time           0.000
     -------------------------------------------------------------
     Instance   Arc          Cell   Delay  Arrival  Required
                                           Time     Time
     -------------------------------------------------------------
     u_q1_reg   CK ^         -      -      0.000    -0.352
     u_q1_reg   CK ^ -> Q ^  DFFX1  0.275  0.275    -0.077
     u_A1       A ^ -> Y ^   OR2X1  0.088  0.363    0.011
     u_out_reg  D ^          DFFX1  0.000  0.363    0.011
     -------------------------------------------------------------
```

Multi-cycle setup exception (×4) on the worst setup path

The reg-to-output path `u_q1_reg/Q → out` is architecturally multi-cycle: the downstream interface only samples this signal every 4 clock cycles

the required time expands from 1 period = 10.000 ns to 4 periods = 40.000 ns; after subtracting the 2.000 ns output delay, the tool reports Required = 38.000 ns with a Cycle Adjustment = 30.000 ns. The data path itself is unchanged (Arrival ≈ 0.924 ns), so setup slack improves from ~7.076 ns to 37.076 ns. Because this exception is setup-only, the hold check is unchanged (same-edge) and still meets timing (hold slack ≈ 0.352 ns), ensuring we don't mask any minimum-delay issues.

Add multi-cycle path exception for the hold analysis also for the above path with a path

multiplier of 3.

sdc:

```
create_clock -name clk -period 10 [get_ports clk]

set_input_delay  2 -clock clk [get_ports {a b}]

set_output_delay 2 -clock clk [get_ports out]

set_multicycle_path 4 -setup -from [get_pins u_out_reg/Q ] -to [get_ports out]
set_multicycle_path 3 -hold -from [get_pins u_out_reg/Q ] -to [get_ports out]

set_drive 0 [get_ports {a b clk}]

set_load 0.1 [get_ports out]
```

Worst Setup :

```
#####################################################################
Path 1: MET Late External Delay Assertion
Endpoint:   out         (^) checked with  leading edge of 'clk'
Beginpoint: u_out_reg/Q (v) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
- External Delay                2.000
+ Phase Shift                  10.000
+ Cycle Adjustment             30.000
= Required Time                38.000
- Arrival Time                  0.924
= Slack Time                   37.076
     Clock Rise Edge                     0.000
     + Clock Network Latency (Ideal)     0.000
     = Beginpoint Arrival Time           0.000
      ------------------------------------------------------------
      Instance   Arc          Cell   Delay  Arrival  Required
                                            Time     Time
      ------------------------------------------------------------
      u_out_reg  CK ^         -      -      0.000    37.076
      u_out_reg  CK ^ -> Q v  DFFX1  0.340  0.340    37.416
      u_N3       A v -> Y ^   INVX1  0.584  0.924    38.000
      -          out ^        -      0.000  0.924    38.000
      ------------------------------------------------------------
```

Worst hold:

```
Path 1: MET Hold Check with Pin u_out_reg/CK
Endpoint:   u_out_reg/D (^) checked with  leading edge of 'clk'
Beginpoint: u_q1_reg/Q  (^) triggered by  leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time          0.000
+ Hold                          0.011
+ Phase Shift                   0.000
- Cycle Adjustment              0.000
= Required Time                 0.011
  Arrival Time                  0.363
  Slack Time                    0.352
     Clock Rise Edge                     0.000
     + Clock Network Latency (Ideal)     0.000
     = Beginpoint Arrival Time           0.000
      ------------------------------------------------------------
      Instance   Arc          Cell   Delay  Arrival  Required
                                            Time     Time
      ------------------------------------------------------------
      u_q1_reg   CK ^         -      -      0.000    -0.352
      u_q1_reg   CK ^ -> Q ^  DFFX1  0.275  0.275    -0.077
      u_A1       A ^ -> Y ^   OR2X1  0.088  0.363    0.011
      u_out_reg  D ^          DFFX1  0.000  0.363    0.011
      ------------------------------------------------------------
```

Hold multi-cycle exception (×3) for the same path

Because the setup capture for `u_q1_reg/Q → out` is shifted to k+4 cycles, we also shift the hold reference to the edge just before the capture, i.e., k+3 cycles

This tells STA to check that data launched at cycle *k* remains stable until the k+3 edge (the edge preceding the k+4 setup capture), avoiding false/over-tight hold requirements that would exist if the tool still checked hold at k+1. Functionally, it preserves realistic minimum-delay checking for the multi-cycle interface while relaxing only what's necessary for correctness. Note that this exception applies only to the reg→port path; your reported worst hold path is reg→reg (`u_q1_reg/Q → u_out_reg/D`), so its hold slack (~0.352 ns) and "Cycle Adjustment = 0.000" remain unchanged—as expected.